XENON INTERPRETABILITY

Prompt Confusion Phase 04 Full Checkpoint

April 10, 2026 — behavior, conflict probing, arbitration readouts, exploratory PCA, section attribution, and the first causal follow-up.

BOTTOM LINE

Phase 04 is mechanistically usable. We can reliably detect *whether* a prompt contains a conflict (peak accuracy 92%), and we have a weaker but real signal for *which side is winning* (peak accuracy 71%). A plausible causal intervention target exists at `SETTINGS mean @ layer 24`, but the first causal test was invalid — the model emitted reasoning traces instead of JSON, so we have no causal result yet.

Plain-language summary

We gave Qwen a batch of 288 synthetic trading prompts — half with internally consistent instructions (aligned), half where the `SETTINGS` and `STRATEGY` sections deliberately contradict each other (conflict). Three things came back clean:

1. **Behavior is usable.** Every prompt produced valid structured output. Conflict prompts behave differently from aligned ones, and they don't all collapse to the same answer — exactly what we need for probing.
2. **The model knows when it's conflicted.** A linear probe on the residual stream can tell conflict from aligned prompts at 92% accuracy (layer 36). This is a strong Level-2 representational finding.
3. **The model partially knows which side is winning.** On conflict-only rows, a probe can decode whether the model is favoring `SETTINGS` or `STRATEGY` at 71% accuracy (layer 20). Weaker, but real.

The one thing that *didn't* work: we tried pushing the `SETTINGS` direction into the residual stream to see if it steers the output. All outputs came back invalid because the model started emitting `<think>` traces. That's a harness bug, not a negative result. Next step is rerunning with thinking disabled.

Run Summary

Behavior	Conflict Probe	Arbitration Probe	Attribution
`16474bce`	`62fc89fd`	`6924c2b3`	`7e71cf74`
288 / 288 valid structured outputs.	Aligned vs. conflict residual probe.	Conflict-only: strategy vs. setting.	123 conflict rows, section-local analysis.

All artifacts live under `phase_04/outputs`:

`behavior_audit_16474bceae4e.json` · `analysis_full_prompt_eos_probe/62fc89fd861b` · `analysis_full_prompt_eos_pca/f4c70fcbac6d` · `analysis_conflict_readout_residual/6924c2b36f43` · `analysis_conflict_readout_pca/331de685cbc1` · `analysis_conflict_readout_router/4aeb882aea9d` · `conflict_arbitration_stage1`

1. Behavioral Sanity

The behavior audit checks whether the synthetic prompts produce usable, non-collapsed differences — not benchmark accuracy. We need conflict rows that behave differently from aligned rows, and conflict rows that don't all resolve the same way.

Split	Rows	Valid	Exact Match	→ Strategy	→ Setting
overall	288	100%	51.4%	66.7%	24.0%
aligned	144	100%	70.8%	83.3%	12.5%
conflict	144	100%	31.9%	50.0%	35.4%

Key takeaway

Every row parsed. Aligned and conflict rows separate behaviorally. Conflict rows show a `72 strategy / 51 setting / 21 neither` readout mix — enough diversity for arbitration analysis.

Conflict rows broken out by family:

Family	Rows	Strategy	Setting	Neither
activity_force_observe	36	31	5	0
activity_force_trade	36	15	21	0
trade_size_force_large	36	3	18	15
trade_size_force_small	36	23	7	6

The softest family (`activity_force_trade`) still moves with contextual pressure rather than collapsing — its `balanced` bucket goes `3 strategy / 9 setting`, while `setting_favored` goes `12 / 0` and `strategy_favored` goes `0 / 12`. Arbitration responds to pressure, not noise.

2. Conflict Detection (Aligned vs. Conflict)

Can we read *whether a conflict exists* from the residual stream? Yes — strongly.

Layer	Balanced Accuracy	Mean	Std Dev	Selectivity
0	0.784	0.784	0.038	0.310
20	0.788	0.788	0.024	0.281
24	0.864	0.864	0.037	0.399
28	0.861	0.861	0.021	0.368
36	**0.920**	**0.920**	**0.042**	**0.417**
44	0.837	0.837	0.072	0.354

Level-2 result

Strong linear readout of conflict state from mid-to-late residual stream, peaking at layer 36 with 92% grouped balanced accuracy. The model builds an increasingly clear internal representation of "these instructions contradict each other" as it processes the prompt.

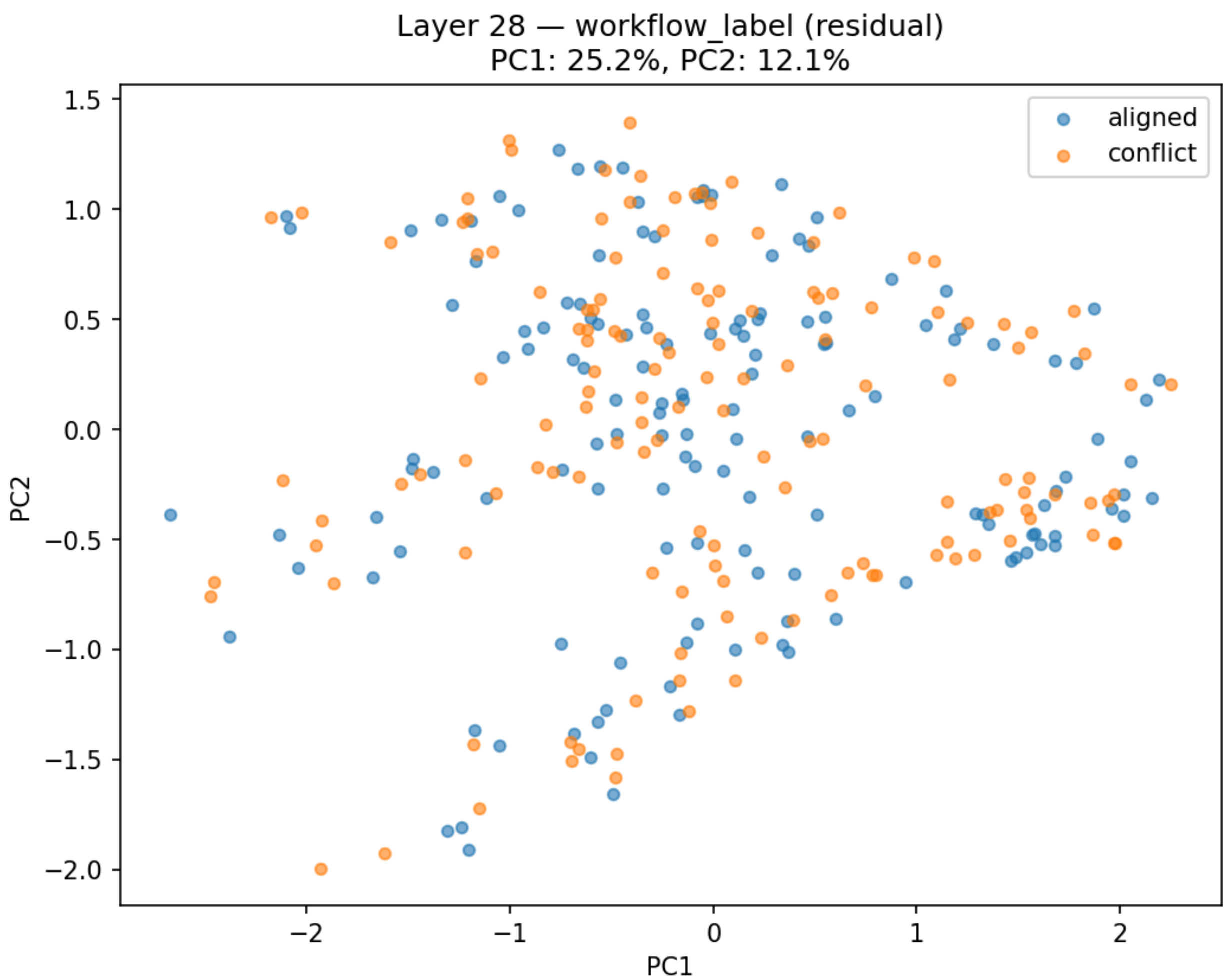

Figure 1: PCA of conflict vs. aligned residual activations at layer 28. Descriptive only, but consistent with the strong linear probe result.

3. Arbitration (Which Side Is Winning?)

The harder question: on conflict-only rows, can we tell whether the model is favoring `SETTINGS` or `STRATEGY`?

Stream	Best Layer	Balanced Accuracy	Std Dev	Note
residual	**`20`**	**`0.712`**	**`0.062`**	Peak arbitration readout.
`residual`	`24`	`0.708`	`0.083`	Nearly tied.
`residual`	`44`	`0.700`	`0.079`	Late residual still carries signal.
`router`	`16`	`0.648`	`0.069`	Best router layer.
`router`	`28`	`0.647`	`0.087`	Comparable.
`router`	`44`	`0.622`	`0.117`	Above baseline, but noisy.

Key takeaway

Arbitration is materially weaker than conflict detection (71% vs. 92%) — expected, since it's a harder question. But it's real: the residual stream carries genuine information about which side of the conflict is winning. Router signal is weaker (65%) but enough to keep in the follow-up stack.

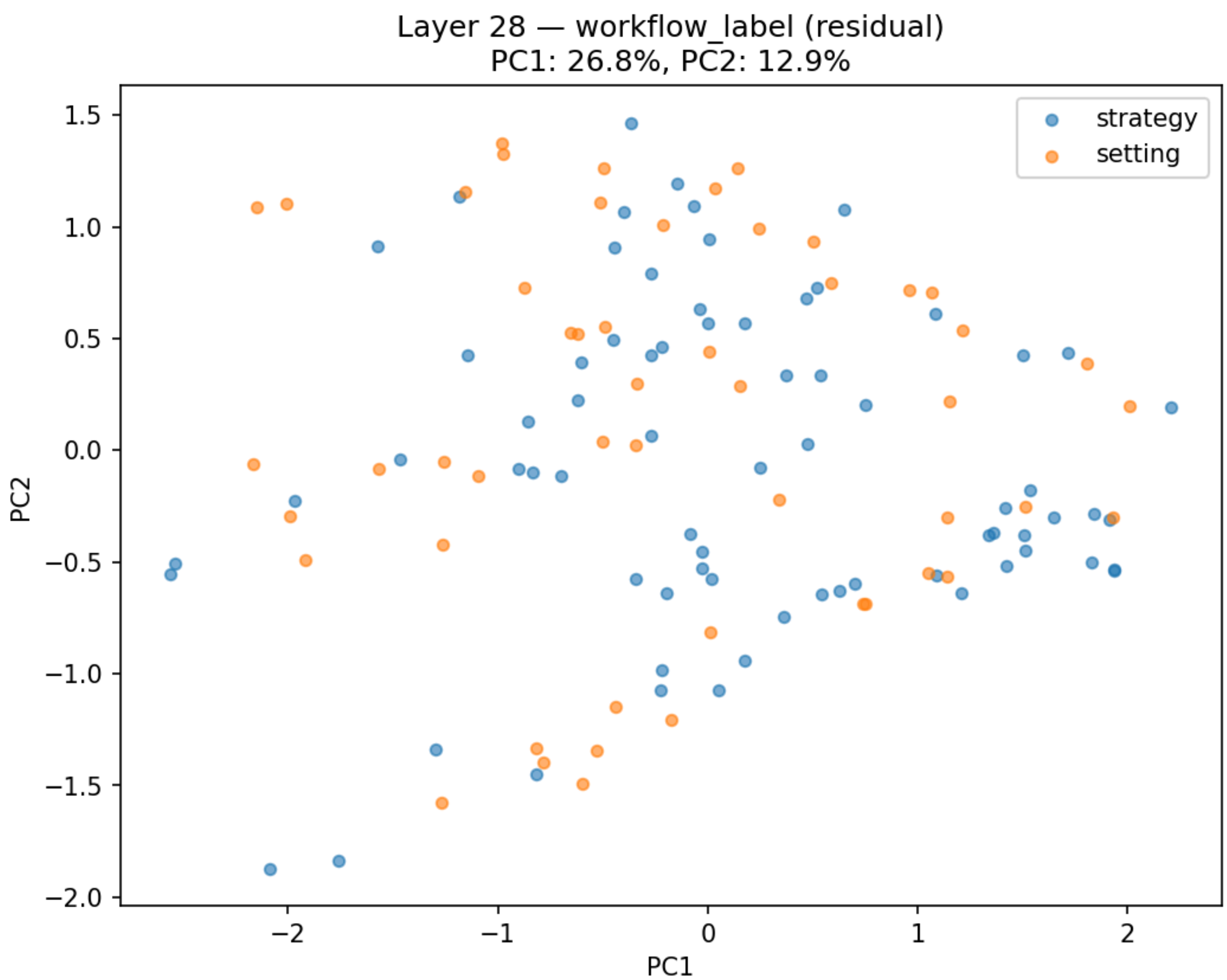

Figure 2: PCA of strategy vs. setting residual activations at layer 28 (conflict rows only).

4. Section Attribution and Causal Status

Where in the prompt is the arbitration signal coming from? The section-attribution pass probed each prompt section independently.

Section	Pooling	Layer	Balanced Acc.	Note
PORTFOLIO	mean	28	0.792	Highest absolute, but confound risk.

MARKET	mean	24	0.784	Strong contextual-pressure confound.
SETTINGS	**mean**	**24**	**0.765**	**Best policy-source readout → selected target.**
SETTINGS	eos	36	0.764	Nearly tied.
STRATEGY	mean	36	0.743	Clear but slightly weaker.

Selected intervention target:

Section: `SETTINGS` · **Pooling:** `mean` · **Layer:** `24` · **Direction norm:** `1.4996`

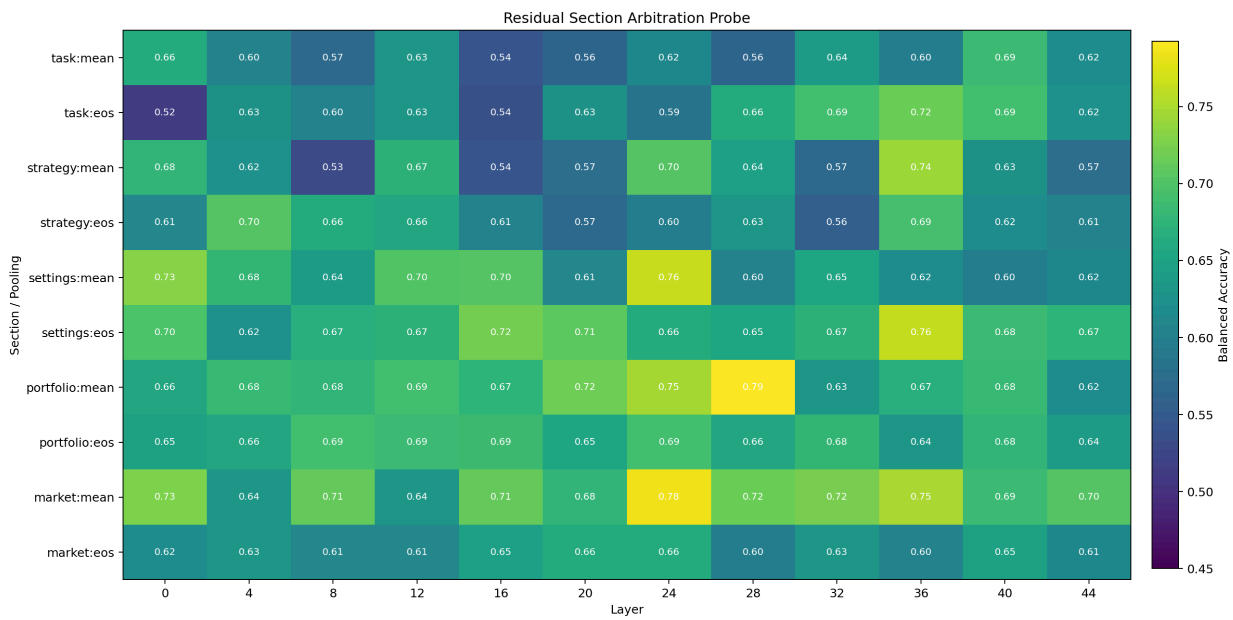

Figure 3: Grouped balanced accuracy by section, pooling, and layer for the conflict-only attribution pass.

Causal Test: Invalid

Not a null result — a harness failure

All three causal conditions (baseline, `setting_push`, `strategy_push`) returned `neither_rate = 1.0`. But the raw outputs show the model emitting `<think> Okay, let's break this down ...` instead of the expected JSON. The intervention may or may not have worked — we can't tell, because the output surface broke. Fix: rerun with thinking disabled.

Two additional caveats:

- Only 25 rows had usable full-span router coverage, so the router section sweep is not yet interpretable.
- `PORTFOLIO` and `MARKET` carry strong arbitration-adjacent signal. Confound control matters in follow-up work.

What We Know vs. What We Don't

ESTABLISHED

- Synthetic prompts are behaviorally usable
- Conflict is strongly represented in the residual stream (92%)
- Arbitration is linearly decodable above baseline (71%)
- Plausible causal target: `SETTINGS mean @ L24`

OPEN

- No valid causal result yet (harness bug)
- Router section attribution too sparse (N=25)
- `PORTFOLIO` / `MARKET` confound not ruled out
- No attention or gradient-based follow-up yet

Next Steps

1. Rerun the causal harness with thinking disabled so outputs match the JSON-only behavioral surface.
2. Keep `SETTINGS mean @ layer 24` as the intervention target unless the rerun disproves it.
3. After a valid causal result, move into attention and attribution follow-up conditioned on conflict-side labels.
4. Maintain explicit confound checks: section length, family identity, and the strong `PORTFOLIO` / `MARKET` readouts.

Local Artifact Index

All paths relative to `projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/`.

- `behavior_audit_16474bceae4e.json`
- `analysis_full_prompt_eos_probe/62fc89fd861b/results.json`
- `analysis_full_prompt_eos_pca/f4c70fcbac6d/pca_layer28_workflow_label.png`
- `analysis_conflict_readout_residual/6924c2b36f43/results.json`
- `analysis_conflict_readout_pca/331de685cbc1/pca_layer28_workflow_label.png`
- `analysis_conflict_readout_router/4aeb882aea9d/results.json`
- `conflict_arbitration_stage1/section_attribution/summary.json`
- `conflict_arbitration_stage1/section_attribution/section_probe_results.parquet`
- `conflict_arbitration_stage1/causal_check/summary.json`
- `conflict_arbitration_stage1/causal_check/row_level.parquet`

Prompt Confusion Phase 04 Full Checkpoint — generated from local outputs under `phase_04/outputs`, including behavior run `16474bce`, conflict probe `62fc89fd`, arbitration probes `6924c2b3` and `4aeb882e`, PCA runs `f4c70fca` and `331de685`, and section-attribution checkpoint `conflict_arbitration_stage1`.